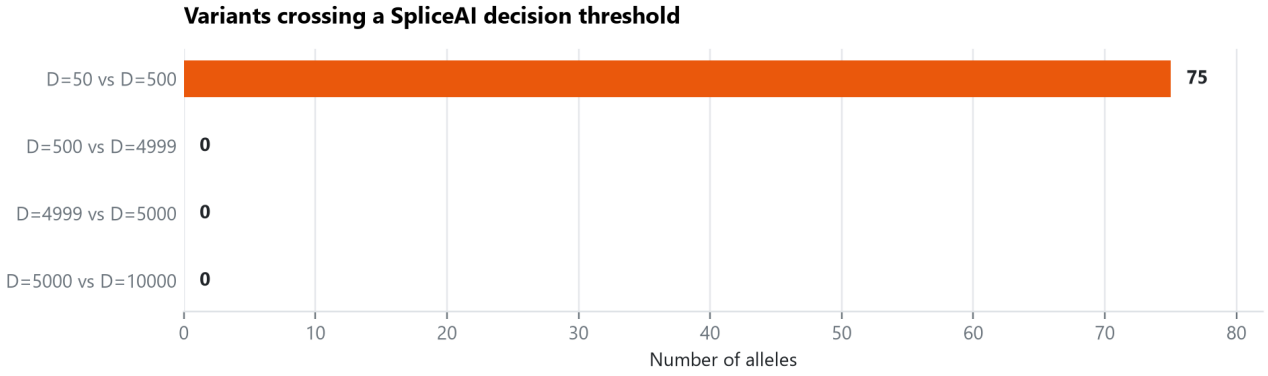
SpliceAI search-distance experiment

Completed stratified BRCA1/2 run using the model, reference sequence and transcript annotation pinned by ARIANE. Results dated 17 September 2026.

4,047	0	75	0
alleles completed	failed evaluations	band changes at D=50 vs D=500	band changes at D=500 vs D=4999

Result

D=50 excluded splice signals that were visible at D=500. In 75 alleles at 52 distinct loci, the maximum score crossed one of the ARIANE decision thresholds at 0.10 or 0.20. No threshold-band change was observed between D=500 and D=4999, or among D=4999, D=5000 and D=10000, in this cohort.

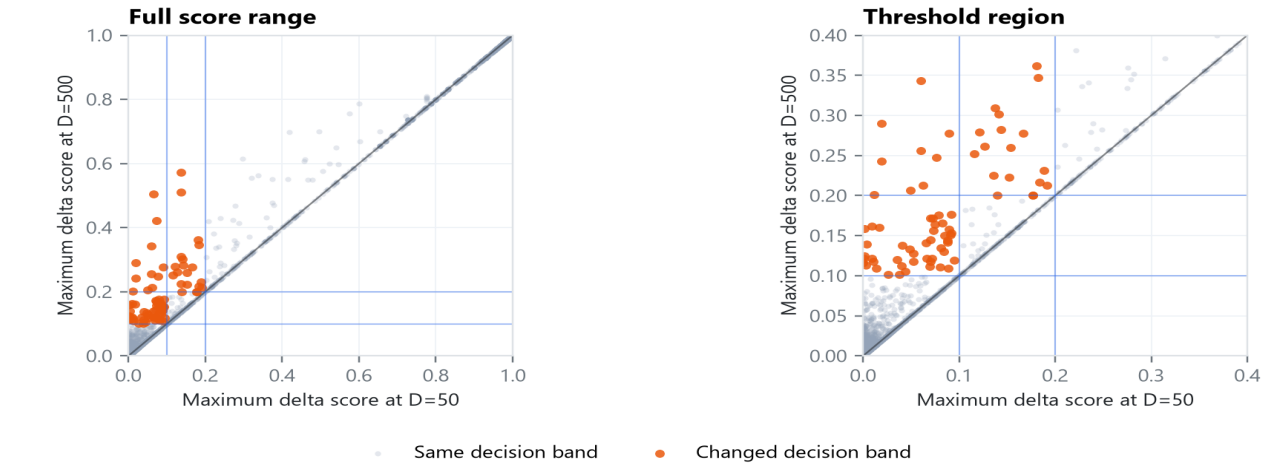


The cohort is an engineering validation set with random and targeted cases. These counts describe the tested cohort and are not prevalence estimates.

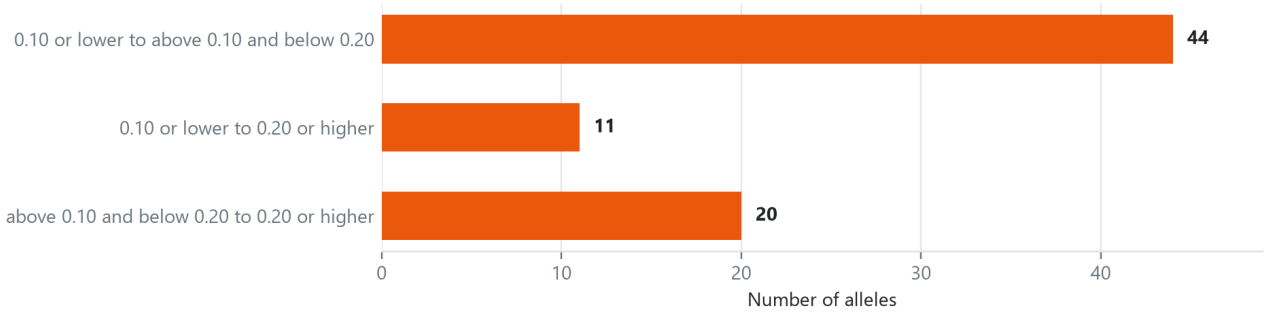
D=50 compared with D=500

Each point represents one tested allele. Orange points changed decision band after the scored window was expanded. Blue guide lines mark the 0.10 and 0.20 thresholds used by the ARIANE profile.

Effect of expanding the scored window from D=50 to D=500



Threshold transitions observed between D=50 and D=500



52	34 / 41	23 / 51 / 1	74 / 1
distinct loci	BRCA1 / BRCA2 alleles	random / targeted / control	SNVs / deletions

Pairwise results

Comparison	Maximum score changed by more than 1e-6	Decision-band changes	Largest absolute score difference
D=50 vs D=500	2127	75	0.439341
D=500 vs D=4999	1106	0	0.045821
D=4999 vs D=5000	0	0	7.75e-07
D=5000 vs D=10000	0	0	1.79e-07

Experimental design

The run evaluated 4,047 distinct alleles at 1,315 positions across all 98 canonical BRCA1 and BRCA2 exon and intron regions. The set contained 3,919 SNVs, 64 deletions and 64 insertions. It included 882 random SNVs at 294 loci plus targeted splice-boundary, distance-boundary and validation cases. Three alternate alleles at the same locus are correlated.

Every evaluation used the same pinned SpliceAI 1.3.4 implementation, container image, GRCh38 reference and GENCODE v49 basic reference transcripts. The D=4999 results in the extension reproduced the saved base run. All 4,047 cases completed and no failed evaluation was counted as agreement.

Interpretation

D=50 is not sufficient for this BRCA1/2 validation set. D=500 and D=4999 gave the same threshold bands in all tested cases, although 1,106 maximum scores differed by more than 1e-6 and the largest difference was 0.045821. The absence of band changes is therefore an empirical result, not proof that the settings are mathematically equivalent for every possible variant.

D=4999, D=5000 and D=10000 produced no threshold-band changes. Their largest observed maximum-score difference was 7.75e-7, below the prespecified 1e-6 numerical tolerance.

Production decision

No ARIANE configuration or classification rule is changed by this experiment. The production profile remains pinned to D=10000, as recorded in data/spliceai/enigma_v1_2_spliceai_profile.json.

Source records: data/spliceai/build/distance-stratified and data/spliceai/build/distance-defaults-server-v2. Analysis tolerance: 1e-6. Decision bands: 0.10 or lower, above 0.10 and below 0.20, and 0.20 or higher.